

1. A particle P of mass 0.3 kg is moving with velocity $5\mathbf{i} \text{ m s}^{-1}$

The particle receives an impulse $\mathbf{I} \text{ N s}$.

Immediately after receiving the impulse, the velocity of P is $(7\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$

(a) Find the magnitude of $\mathbf{I}$

(4)

(b) Find the angle between the direction of $\mathbf{I}$ and the direction of motion of P immediately before receiving the impulse.

(3)

2. [In this question, the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a horizontal plane.]

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A particle P is moving on a smooth horizontal plane.

At time t seconds ($t \geq 0$), the position vector of P , relative to a fixed point O , is $\mathbf{r}$ metres and the velocity of P is $\mathbf{v}$ ms^{-1} where

$$\mathbf{v} = (4t^2 - 5t)\mathbf{i} + (-10t - 12)\mathbf{j}$$

When $t = 0$, $\mathbf{r} = 2\mathbf{i} + 6\mathbf{j}$

(a) Find $\mathbf{r}$ when $t = 2$

(4)

When $t = T$ particle P is moving in the direction of the vector $\mathbf{i} - 2\mathbf{j}$

(b) Find the value of T

(3)

(c) Find the exact magnitude of the acceleration of P when $t = 2.5$

(3)

3.

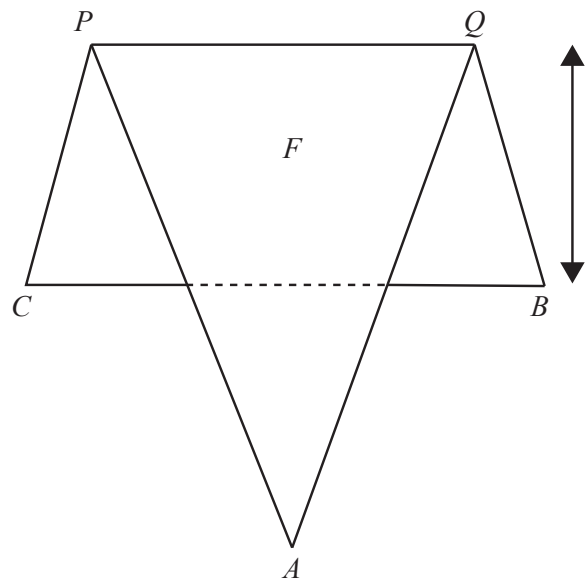
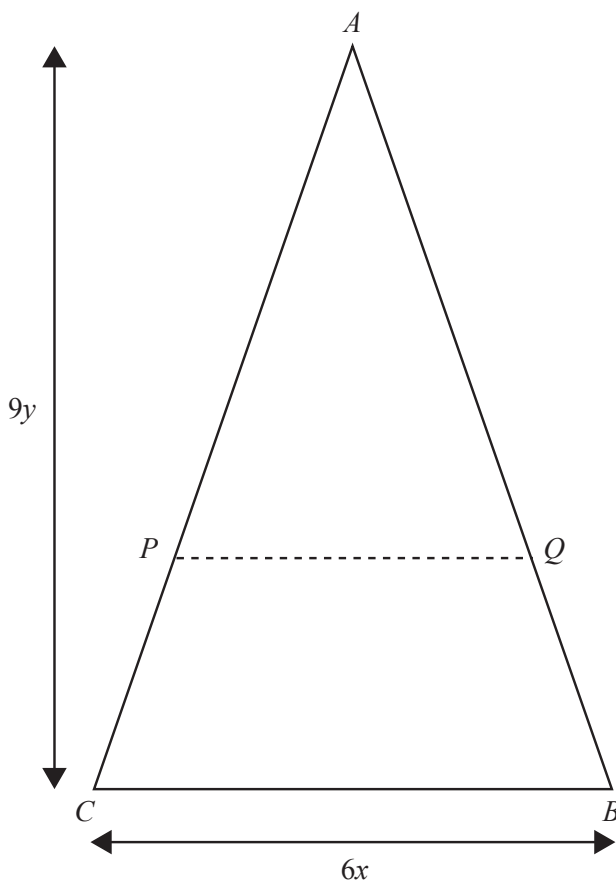


Figure 1

The uniform triangular lamina ABC , shown in Figure 1, has height $9y$, base $BC = 6x$, and $AB = AC$

The points P and Q are such that $AP : PC = AQ : QB = 2 : 1$

The lamina is folded along PQ to form the folded lamina F

The distance of the centre of mass of F from PQ is d

(a) Show that $d = \frac{16}{9}y$ (5)

The folded lamina is suspended from P and hangs freely in equilibrium with PQ at an angle α to the downward vertical.

Given that $\tan \alpha = \frac{64}{81}$

(b) find x in terms of y (3)

4. A particle P of mass $3m$ and a particle Q of mass $5m$ are moving towards each other along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately **before** the collision, the speed of P is u and the speed of Q is ku .

Immediately **after** the collision, the speed of P is $2v$ and the speed of Q is v .

The direction of motion of each particle is reversed by the collision.

In the collision, P receives an impulse of magnitude $15mv$.

- (a) Show that $u = 3v$.

(3)

- (b) Find the value of k .

(3)

The coefficient of restitution between P and Q is e .

- (c) Find the value of e .

(3)

The total kinetic energy lost in the collision is λmv^2

- (d) Find the value of λ .

(3)

5.

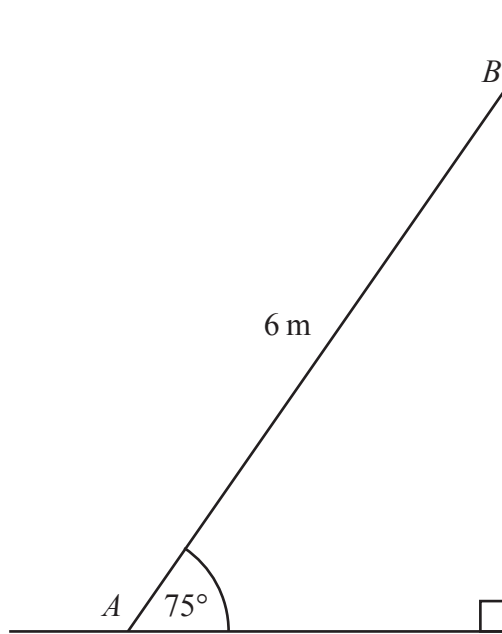


Figure 2

A uniform beam AB , of mass 15 kg and length 6 m, rests with end A on rough horizontal ground. The end B of the beam rests against a rough vertical wall.

The beam is inclined at 75° to the ground, as shown in Figure 2.

The coefficient of friction between the beam and the wall is 0.2

The coefficient of friction between the beam and the ground is μ

The beam is modelled as a uniform rod which lies in a vertical plane perpendicular to the wall.

The beam rests in limiting equilibrium.

(a) Find the magnitude of the normal reaction between the beam and the wall at B .

(5)

(b) Find the value of μ

(6)

6. A van of mass 900 kg is moving along a straight horizontal road.

The resistance to the motion of the van is modelled as a constant force of magnitude 600 N.

The engine of the van is working at a constant rate of 24 kW.

At the instant when the speed of the van is $V \text{ ms}^{-1}$, the acceleration of the van is 2 ms^{-2}

- (a) Find the value of V

(4)

Later on, the van is towing a trailer of mass 700 kg up a straight road inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{14}$

The trailer is attached to the van by a towbar, as shown in Figure 3.

The towbar is parallel to the direction of motion of the van and the trailer.

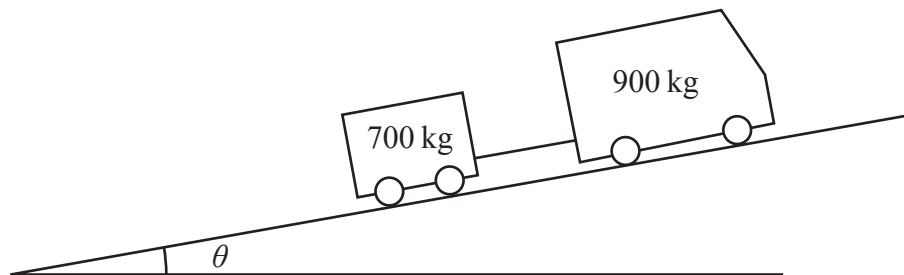


Figure 3

The resistance to the motion of the van from non-gravitational forces is modelled as a constant force of magnitude 600 N.

The resistance to the motion of the trailer from non-gravitational forces is modelled as a constant force of magnitude 550 N.

The towbar is modelled as a light rod.

The engine of the van is working at a constant rate of 24 kW.

- (b) Find the acceleration of the van at the instant when the van and the trailer are moving with speed 8 ms^{-1}

(4)

At the instant when the van and the trailer are moving up the road at 9 ms^{-1} , the towbar breaks. The trailer continues to move in a straight line up the road until it comes to instantaneous rest.

The distance moved by the trailer as it slows from a speed of 9 ms^{-1} to instantaneous rest is d metres.

- (c) Use the work-energy principle to find the value of d .

(4)

7. [In this question, the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane with $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upwards.]

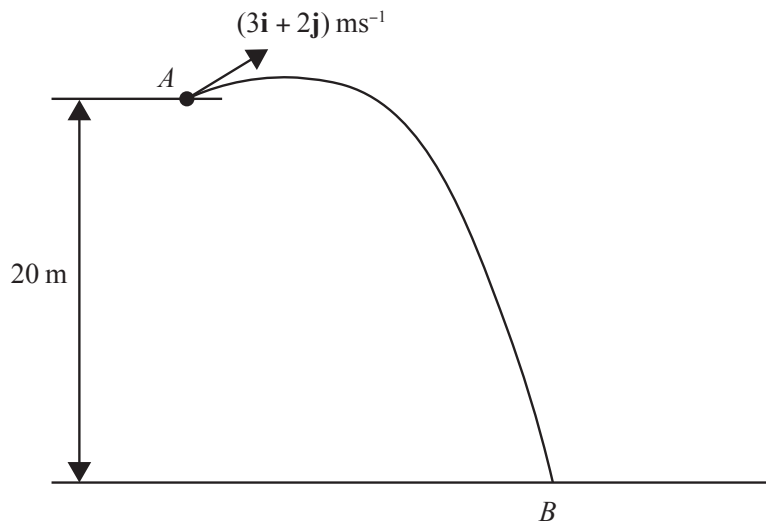


Figure 4

A small ball is projected with velocity $(3\mathbf{i} + 2\mathbf{j}) \text{ ms}^{-1}$ from the fixed point A .

The point A is 20 m above horizontal ground.

The ball hits the ground at the point B , as shown in Figure 4.

The ball is modelled as a particle moving freely under gravity.

- (a) By considering energy, find the speed of the ball at the instant immediately before it hits the ground. (3)
- (b) Find the direction of motion of the ball at the instant immediately before it hits the ground. (3)
- (c) Find the time taken for the ball to travel from A to B . (3)
- At the instant when the direction of motion of the ball is perpendicular to $(3\mathbf{i} + 2\mathbf{j})$ the ball is h metres above the ground.
- (d) Find the value of h . (6)